(i) Let $f(x) = x - \ln\big(1+x\big)$

$$f(0) = 0, \qquad f'(x) = 1 - \frac{1}{1+x} = \frac{x}{1+x} > 0$$

$$\therefore x - \ln\big(1+x\big) > 0 \implies x > \ln\big(1+x\big) \qquad (x > 0)$$

When $x = \dfrac{1}{k}$

$$\frac{1}{k} > \ln\big(1+\frac{1}{k}\big) = \ln\frac{k+1}{k} = \ln\big(k+1\big) - \ln\big(k\big)$$

$$\sum_{k=1}^{n}\frac{1}{k} > \sum_{k=1}^{n}\Big(\ln(k+1) - \ln(k)\Big) = \ln(n+1)$$

(ii) Let $g(x) = x + \ln\big(1-x\big)$

$$g(0) = 0, \qquad g'(x) = 1 - \frac{1}{1-x} = -\frac{x}{1-x} < 0$$

$$\therefore x + \ln(1-x) < 0 \implies x < -\ln\big(1-x\big) \qquad (0 < x < 1)$$

When $x = \dfrac{1}{k^2}$, where $k \geq 2$

$$\frac{1}{k^2} < -\ln\big(1-\frac{1}{k^2}\big) = \ln(\frac{k^2}{k^2-1}) = 2\ln k - \ln(k-1) - \ln(k+1)$$

$$\begin{aligned}\sum_{k=2}^{n}\frac{1}{k^2} &< \sum_{k=2}^{n}\big(2\ln k - \ln(k-1) - \ln(k+1)\big)\\ &= \ln 2 + \ln n - \ln(n+1)\end{aligned}$$

When $n \to \infty$

$$\sum_{k=2}^{\infty}\frac{1}{k^2} < \ln 2$$

$$\implies \sum_{k=1}^{\infty}\frac{1}{k^2} = 1 + \sum_{k=2}^{\infty}\frac{1}{k^2} < 1 + \ln 2$$

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